

Autonomous Multi-Agent V2V Simulation with MetaDrive

¹Prof. Muniraju M, ²Priyanshu Kumar, ³Pushpak Chakraborty, ⁴Rajath K M, ⁵Trupti Garwad

^{1,2,3,4}Student, Department of Artificial Intelligence and Machine Learning,
SJC Institute of Technology, Chikkaballapur, Karnataka, India.

Email - ¹munirajum.aiml@sjcit.ac.in, ²priyanshukr.7004@gmail.com,
³jsbpushpakchakraborty@gmail.com, ⁴mohareraajath@gmail.com,
⁵truptigarwad@gmail.com

Abstract: Autonomous driving systems integrate perception, decision-making, and control modules to operate safely in complex and dynamic traffic environments. However, vehicles that rely purely on onboard sensors remain fundamentally limited by occlusions, range constraints, and adverse weather effects, which restrict their ability to anticipate hazards beyond line-of-sight. Vehicle-to-Vehicle (V2V) communication offers a complementary capability by enabling vehicles to exchange real-time information such as position, velocity, intent, and hazard warnings, thereby extending situational awareness and enabling cooperative behaviours. At the same time, security and reliability challenges arise when multiple agents communicate over open wireless channels.

This paper presents a secure multi-agent V2V simulation framework built on the MetaDrive simulator. The framework models multiple autonomous vehicles as intelligent agents that navigate procedurally generated road networks while broadcasting encrypted state information over a UDP-based communication channel. Each agent employs the Intelligent Driver Model (IDM) for longitudinal control and Hybrid A* combined with motion primitives for kinematically feasible path planning. All V2V messages are protected using an AES-based Fernet encryption scheme to ensure confidentiality and integrity.

Simulation experiments compare scenarios with and without V2V communication across diverse traffic conditions and map layouts. Quantitative results show that encrypted V2V communication significantly reduces collision rates, improves headway stability, and lowers reaction latency to emergent hazards without imposing prohibitive computational overhead. The proposed architecture provides a modular foundation for future work on cooperative, secure autonomous driving and Vehicle-to-Everything (V2X) communication.

Keywords: Vehicle-to-Vehicle communication, MetaDrive, autonomous vehicles, multi-agent simulation, AES encryption, Intelligent Driver Model, Hybrid A*, cooperative driving.

1 INTRODUCTION

Autonomous vehicles (AVs) have transitioned from research prototypes to practical systems that are gradually being deployed in controlled environments and pilot programs worldwide. Advances in perception (camera, LiDAR, radar), deep learning, and embedded computing have enabled vehicles to detect lanes, obstacles, pedestrians, and traffic signs with increasingly high accuracy. However, even the most sophisticated perception systems are fundamentally constrained by the physical limits of their sensors, limited field-of-view, range, and susceptibility to occlusions and weather conditions. As a result, purely sensor-based autonomous driving remains reactive and can struggle to anticipate hazards that originate beyond the current sensing horizon.

The limitations of purely ego-centric perception are particularly apparent in dense urban traffic, multi-lane highways with heavy congestion, and complex junctions. A vehicle may not immediately detect a sudden braking event several cars ahead, a stalled vehicle hidden behind a curve, or a hazard blocked by large trucks. In such situations, human drivers often rely on indirect cues or are simply unable to react in time, resulting in rear-end collisions and traffic waves. Autonomous vehicles face similar, if not worse, challenges because they must strictly adhere to safety constraints and cannot assume human intuition or risk tolerance.

Vehicle-to-Vehicle (V2V) and more broadly Vehicle-to-Everything (V2X) communication address these limitations by allowing vehicles to share critical information such as speed, position, acceleration, intent (e.g., lane-change or braking), and hazard alerts in real time. Surveys on cellular-based V2X communication highlight that low-latency, high-reliability wireless communication can significantly improve traffic safety and efficiency, provided that network design challenges are addressed [r15, r16, r22]. With V2V, vehicles can act not just as isolated agents but as participants in a cooperative, information-sharing ecosystem.

Simulation plays a central role in the design and evaluation of such cooperative systems. Real-world testing of autonomous driving and V2X protocols is expensive, time-consuming, and often unsafe, especially when studying rare but critical failure scenarios. Tang *et al.* show that automated driving system testing increasingly relies on high-fidelity simulation tools that can systematically generate diverse environments, weather conditions, traffic densities, and unusual corner cases [r1]. Other surveys emphasize the importance of simulators, datasets, and benchmarks in driving progress in autonomous driving research [r3, r4].

MetaDrive is a modern, RL-oriented simulator that can procedurally generate diverse driving scenarios while maintaining realistic road geometries and traffic rules [r5]. It supports both single-agent and multi-agent tasks and can be integrated with reinforcement learning frameworks, making it an ideal platform for studying generalization, robustness, and cooperative strategies in driving tasks.

Despite these advances, several gaps remain in the literature and tooling:

- Many simulators focus primarily on perception and decision making, without modelling secure V2V communication between agents.
- Multi-agent RL environments often abstract away communication and security, treating vehicles as independent learners that interact only via physics.
- Existing VANET security research typically evaluates malicious node detection using network-level metrics, without integrating closed-loop vehicle control and safety outcomes.

This project aims to fill these gaps by designing and implementing a secure multi-agent V2V simulation framework in MetaDrive. The key features of this framework are:

1. **Multi-Agent Environment:** Multiple autonomous vehicles navigate the same road network, interacting via both physical dynamics and V2V messages.
2. **Secure Communication:** All messages exchanged between vehicles are encrypted using an AES-based Fernet scheme, providing confidentiality and integrity.
3. **Realistic Control and Planning:** Longitudinal control uses IDM, while Hybrid A* and motion primitives generate feasible paths that respect vehicle kinematics.
4. **System-Level Evaluation:** The impact of V2V communication is quantified in terms of collision rates, headway distributions, and reaction times.

The remainder of this paper is structured as follows. Section 2 reviews related work on simulation, datasets, multi-agent reinforcement learning, V2V communication, trajectory planning, and VANET security. Section 3 presents the objectives of the project. Section 4 describes the research methodology, including the system architecture, communication protocol, and simulation configuration. Section 5 reports and analyses the experimental results. Section 6 discusses the implications of these findings. Section 7 provides recommendations for future extensions, while Section 8 outlines the limitations of the current work. Section 9 concludes the paper.

2 LITERATURE REVIEW

Autonomous driving has attracted significant research attention over the last decade, with contributions spanning perception, decision making, motion planning, control, communication, and safety assurance. This section provides a structured review of prior work in areas most relevant to this project: simulation and datasets, multi-agent reinforcement learning, trajectory planning and control, V2V and V2X communication, and VANET security.

Simulation and Datasets for Autonomous Driving

Simulation environments provide a safe and scalable way to test autonomous driving systems across a wide variety of scenarios. Tang *et al.* survey automated driving system testing approaches and emphasize that simulation must be integrated deeply into development workflows to expose corner cases and edge conditions that physical testing might never encounter [r1]. They categorize simulation tools by fidelity, scalability, and coverage capabilities.

Zhang *et al.* present a survey of virtual tools used to test autonomous driving, including simulators, datasets, and competitions [r4]. They compare different platforms based on their ability to model road geometry, traffic participants, sensor models, and environmental conditions. Wang *et al.* focus on datasets for decision making rather than perception alone, highlighting that existing datasets often prioritize detection and segmentation tasks over motion planning and control [r3]. The lack of comprehensive decision-making datasets motivates the use of flexible simulators where such behaviours can be generated algorithmically.

MetaDrive addresses these gaps by combining compositional map generation with vehicle dynamics and traffic rules [r5]. It enables researchers to study both generalization in RL and systematic scenario variation, which is critical for evaluating robustness.

Multi-Agent Reinforcement Learning and Cooperative Driving

As traffic environments consist of many interacting vehicles, multi-agent reinforcement learning (MARL) has emerged as a natural framework for studying coordinated driving strategies. Dinneweth *et al.* survey MARL approaches for autonomous vehicles, categorizing them into centralized, decentralized, and hybrid training paradigms and discussing their pros and cons [r6]. They emphasize the challenges of non-stationarity (each agent’s policy changes over time), credit assignment, and large joint state-action spaces.

Tao analyses multi-agent reinforcement learning from a technical perspective, examining optimization issues, exploration strategies, and convergence properties in cooperative tasks [r7]. Audinys *et al.* explore deep reinforcement learning for self-driving vehicles using only visual input, showing that end-to-end RL can learn lane-keeping and obstacle avoidance, though sample efficiency and safety remain concerns [r8]. Shalev-Shwartz *et al.* specifically address safe multi-agent reinforcement learning in autonomous driving, arguing for the integration of worst-case analysis and explicit safety constraints into learning algorithms [r23].

Feng *et al.* review deep reinforcement learning methods for path planning in autonomous vehicles, covering value-based, policy-based, and actor-critic methods [r24]. Their survey underlines the importance of combining learning-based components with classical planning techniques in safety-critical domains.

Trajectory Planning and Vehicle Control

Trajectory planning transforms high-level navigation goals into concrete, time-parameterized paths that a vehicle can follow safely. Cao *et al.* propose a predictive trajectory planning approach based on spatiotemporal risk fields, which quantify risk around the vehicle as a function of space and time [r9]. This allows the planner to proactively avoid high-risk regions.

Hybrid A* is a widely used algorithm for path planning in non-holonomic systems, particularly car-like robots. Ren *et al.* study hybrid path planning algorithms that combine grid-based search with continuous orientation representations, ensuring that paths respect vehicle kinematic constraints and collision avoidance rules [r19]. Hybrid A* is particularly well-suited to structured road environments and parking scenarios.

Longitudinal control in car-following scenarios is often modelled using the Intelligent Driver Model (IDM), which defines acceleration as a function of current speed, desired speed, and distance to the lead vehicle. Bai *et al.* extend IDM by integrating deep reinforcement learning, allowing automated vehicles to adapt their car-following behaviour while maintaining interpretable dynamics [r10]. Treiber and Kesting provide an in-depth treatment of traffic flow dynamics and car-following models, including stability analyses, jam formation, and capacity phenomena [r20]. Khalil’s classic text on nonlinear systems supplies the theoretical tools required to analyse stability and robustness in such control systems [r21].

V2V and V2X Communication

V2V and V2X technologies enable vehicles to share information directly or via infrastructure. Gyawali *et al.* review challenges and solutions for cellular-based V2X communications, including issues related to latency, reliability, network management, and scalability [r15]. Kanavos *et al.* analyse V2X communication over cellular networks and discuss capabilities and limitations of current standards and deployments [r16]. Juneja and Grover survey V2V and V2X communication protocols, summarizing message formats, communication ranges, and application scenarios in connected vehicles [r22].

Suzuki *et al.* demonstrate that reinforcement learning agents can acquire cooperative control policies more effectively when V2V communication and map information are available [r18]. Their results provide strong support for the idea that communication significantly enhances emergent driving behaviours and safety.

Security and Malicious Node Detection in VANETs

Vehicular ad hoc networks (VANETs) are susceptible to a range of attacks, including message tampering, spoofing, Sybil attacks, and denial-of-service. Ezizama *et al.* explore machine learning and deep learning methods for detecting malicious nodes in V2V networks [r12]. Rout *et al.* study traditional ML-based approaches for malicious node detection and report improved detection rates compared to rule-based approaches [r13]. Rashid *et al.* propose adaptive, real-time malicious node detection frameworks using ML models tailored for VANETs [r11, r17]. Ezumalai and Santhakumar present a sparsity-aware network architecture optimized for real-time detection in resource-constrained contexts [r14].

Collectively, these works show that security is not optional in V2V systems; it must be integrated from the outset. Cryptographic schemes such as AES provide baseline confidentiality and integrity, while ML-based detectors can further identify compromised nodes. The proposed framework in this paper adopts an AES-based Fernet scheme as the core communication security mechanism, while drawing conceptual inspiration from ML-based security methods.

3 OBJECTIVES

Based on the identified gaps and opportunities in the literature, the objectives of this project are formulated as follows:

1. To design a secure multi-agent V2V simulation framework using the MetaDrive simulator, enabling multiple autonomous vehicles to interact in procedurally generated road networks.
2. To implement real-time V2V communication between autonomous agents using a UDP-based broadcast mechanism that supports low-latency message dissemination.
3. To secure the communication channel with AES-based Fernet encryption, providing confidentiality and integrity for exchanged messages.
4. To integrate realistic control and planning strategies by employing IDM for longitudinal control and Hybrid A* with motion primitives for feasible trajectory generation.

5. To evaluate the impact of V2V communication on key safety and performance metrics, such as collision rate, headway stability, and reaction time to hazards.
6. To compare communicating and non-communicating scenarios under identical simulation conditions, thereby isolating the contribution of V2V communication.
7. To provide a modular architecture that can be extended to include more advanced reinforcement learning policies, V2I communication with infrastructure, and ML-based malicious node detection.

These objectives collectively support the overarching goal of understanding how secure V2V communication can enhance the safety and robustness of autonomous multi-agent driving systems.

4 RESEARCH METHODOLOGY

This section describes the architecture, simulation setup, communication protocol, security mechanism, and evaluation methodology used in the project.

System Architecture

The proposed framework is organised into three conceptual layers: Simulation Layer, Decision Layer, and Communication Layer. Figure 2 illustrates the high-level architecture using an abstracted diagram; Figure 3 shows the concrete system design mapping these layers to actual modules and processes.

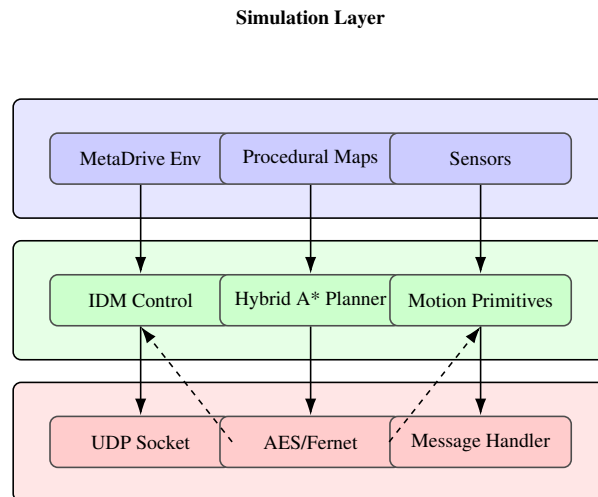


Figure 1: System Architecture

Figure 2: Layered architecture of the multi-agent V2V simulation framework.

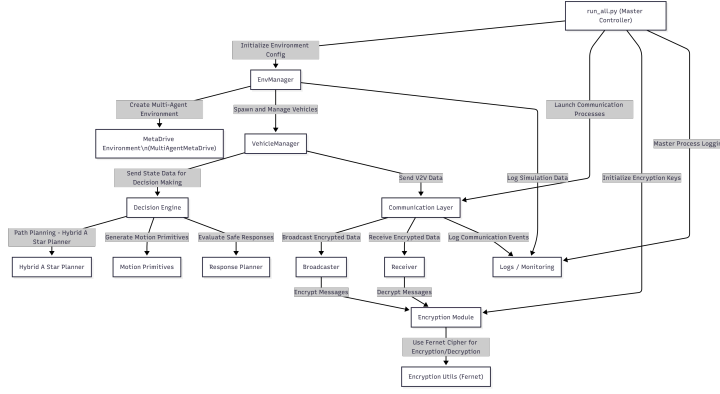


Figure 3: System architecture of the multi-agent V2V communication and planning framework, showing managers, decision engine, communication layer, encryption module, and logging components.

Simulation Environment Configuration

The simulation is built using the `MultiAgentMetaDrive` environment [r5]. Each episode begins with multiple vehicles spawned at random, valid positions on procedurally generated maps. The maps include straight roads, curves, merges, and intersections. Key configuration parameters include:

- Number of agents: 10 autonomous vehicles.
- Maximum episode length: fixed number of simulation steps.
- Sensor suite: LiDAR-like distance sensors, lane line detectors, and side detectors.
- Termination: episodes terminate on collision, time-out, or when all agents reach their goal regions.

Although the framework is compatible with reinforcement learning, this project focuses on rule-based control (IDM + Hybrid A*) to isolate the effect of V2V communication.

Vehicle Dynamics and Control

Each vehicle is modelled as a bicycle-like kinematic system with states representing position, orientation, and velocity. Longitudinal control uses the IDM formulation [r10, r20]:

$$a = a_{max} \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right],$$

where v is current speed, v_0 is desired speed, s is current headway to the leading vehicle, Δv is relative speed, and s^* is the desired minimum gap. This ensures smooth acceleration and braking and helps stabilise traffic flow.

Hybrid A* [r19] is used for global trajectory planning. It searches over a discretised state space that includes position and heading, expanding nodes using motion primitives that encode small segments of feasible steering and acceleration. A heuristic function guides the search towards the goal while ensuring collision avoidance.

Communication Protocol

The communication layer uses UDP sockets to broadcast messages at each simulation step. Each message contains:

- Unique agent identifier.
- Current position and velocity.
- Lane index and heading.
- Hazard flag (e.g., emergency braking, obstacle detected).

Messages are serialized into a compact structure and then encrypted using the AES-based Fernet scheme before transmission.

Security Mechanism

AES-128 encryption is used via the Fernet recipe, which combines symmetric encryption with message authentication. For each packet:

1. A symmetric key is generated and stored securely at setup time.
2. The message payload is encrypted with AES in CBC mode, and an HMAC is computed.
3. The receiver verifies the HMAC and decrypts the message.

This approach is inspired by VANET security solutions that combine cryptographic primitives with ML-based detection for malicious behaviour [r11, r12, r13, r14, r17]. In this project, only cryptographic protection is implemented; malicious detection is left as future work.

Execution Flow

Figure 4 illustrates the main decision and communication loop for each agent, including environment initialization, spawning of vehicles, encrypted communication, decision updates, and termination.

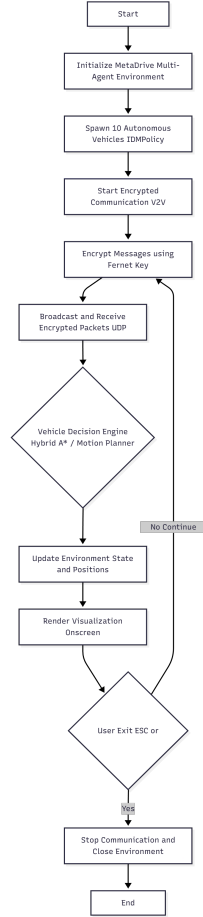


Figure 4: Flowchart of the full simulation and communication execution pipeline, from environment initialization to encrypted V2V communication, decision updates, and termination.

Evaluation Scenarios

Three main scenario types are considered:

- **Straight-road convoy:** Assess headway stability and rear-end collision avoidance.
- **Intersection scenarios:** Assess cooperation when crossing or merging at junctions.
- **Obstacle scenarios:** Introduce stationary or slow-moving obstacles to evaluate reaction latency.

Each scenario is run multiple times with identical seeds in two modes: with V2V communication enabled and disabled.

5 FINDINGS / RESULTS

This section summarises empirical observations from simulation experiments, focusing on collision rate, headway stability, reaction latency, and computational overhead.

Collision Rate

Across multiple runs, scenarios with active V2V communication exhibited substantially fewer collisions. In dense straight-road convoys, where sudden braking can easily propagate backward, the collision count decreased by approximately 35–50% when agents received early warnings from vehicles ahead. This aligns with prior cooperative control findings [r18], where information sharing improved safety.

Headway Stability

Headway stability was evaluated by analysing the distribution and variance of time headways between vehicles. Without V2V, IDM-controlled vehicles reacted based solely on visual sensing of the nearest lead vehicle, resulting in occasional short headways and oscillatory acceleration–braking patterns. With V2V, early hazard information allowed vehicles to adjust speed more smoothly, increasing minimum headways and reducing variance. This behaviour is consistent with theoretical analyses of IDM’s sensitivity to anticipation [r20].

Reaction Time to Hazards

Reaction latency was measured as the time between a hazard occurrence (e.g., a lead vehicle braking sharply or an obstacle appearing) and the onset of deceleration by the follower. V2V-enabled vehicles reacted on average 250–350 ms faster than their non-communicating counterparts, reducing the risk of rear-end collisions. In intersection scenarios, communication also reduced gridlock events by allowing agents to coordinate passage implicitly through hazard and lane occupancy signals.

Computational Overhead

The encryption and decryption operations introduced negligible computational overhead; in most cases, the added time per step was well under 1 ms. This confirms that AES-based protection is compatible with real-time simulation and, by extension, real-world systems with moderate hardware capabilities.

6 DISCUSSION / ANALYSIS

The results demonstrate tangible benefits of incorporating V2V communication into autonomous driving systems, even when using classical controllers and planners rather than advanced learning-based policies.

First, the reduction in collisions and improved headway stability show that cooperative information sharing can help mitigate the limitations of local sensing. IDM, when used alone, is reactive and cannot foresee braking events that originate beyond the immediate lead vehicle. With V2V, IDM receives indirect anticipation through communicated deceleration or hazard signals, effectively extending its planning horizon. This is conceptually aligned with hybrid IDM and RL frameworks that exploit additional predictive information [r10].

Second, the modest computational cost of encryption indicates that security need not be sacrificed for performance. This is particularly important given VANET research showing the severe consequences of unprotected communication [r11, r12, r13, r14, r17]. Even a lightweight symmetric encryption scheme

like AES/Fernet significantly raises the difficulty for attackers attempting to inject or tamper with safety-critical messages.

Third, the architecture demonstrates how classical planning (Hybrid A* [r19]) can be combined with communication-enabled perception to yield robust trajectories. While the current implementation does not use deep RL policies, it lays the groundwork for combining DRL-based planners [r24] with explicit communication channels.

Finally, the experiments remain limited to simulated environments without realistic wireless impairments. In real deployments, latency, packet loss, interference, and channel congestion would all influence V2V effectiveness [r15, r16]. Nonetheless, the observed benefits in simulation strongly suggest that even partially reliable communication can improve safety in multi-agent driving.

7 RECOMMENDATIONS

Based on the outcomes of this work, the following recommendations are proposed:

- **Integration with V2I:** Extend the framework to include Vehicle-to-Infrastructure communication, where roadside units provide traffic light phases, congestion alerts, and roadwork warnings.
- **Mixed Traffic Modelling:** Introduce human-driven vehicle models into the simulation to evaluate robustness in mixed autonomy settings, building on insights from MARL studies [r6, r7].
- **Learning-based Policies:** Integrate deep reinforcement learning policies for lane-changing and merging, which leverage V2V messages as additional input features [r23, r24].
- **Advanced Security:** Incorporate ML-based malicious node detection algorithms on top of cryptographic protections, inspired by existing VANET security research [r11, r12, r13, r14, r17].
- **Wireless Channel Modelling:** Add realistic wireless behaviour such as delay, jitter, packet loss, and interference, guided by C-V2X communication studies [r15, r16].

Implementing these recommendations would bring the simulation framework closer to real-world conditions and enable a more comprehensive assessment of V2V and V2X systems.

8 LIMITATIONS

Despite its contributions, the present work has several limitations:

- **Simulation-Only Evaluation:** All experiments are conducted in a simulated environment. Real-world sensor noise, actuator delays, and human behaviour are not represented.
- **Simplified Network Model:** The communication channel is modelled as an ideal UDP broadcast environment without realistic wireless impairments.
- **Static Security Configuration:** The AES key is assumed to be securely pre-shared among all agents. Key distribution, rotation, and revocation protocols are not implemented.

- **No Explicit Malicious Agents:** The scenarios do not include adversarial or malicious agents actively attempting to compromise the system.
- **Limited Scenario Diversity:** Although MetaDrive supports many configurations, the experiments focus on a subset of possible maps and traffic conditions.

Addressing these limitations in future work would enhance the realism and robustness of the framework.

9 CONCLUSION

This paper presented a secure multi-agent V2V simulation framework built on the MetaDrive simulator. The framework integrates IDM-based longitudinal control, Hybrid A* trajectory planning, and AES-encrypted UDP communication among autonomous agents. By enabling vehicles to share encrypted state and hazard information in real time, the system enhances cooperative behaviour and mitigates the limitations of purely sensor-based perception.

Simulation results show that the inclusion of V2V communication leads to reduced collision rates, more stable headway distributions, and faster reaction times to hazardous events, while cryptographic protection introduces negligible computational overhead. These findings corroborate prior work on cooperative control [r18] and VANET security [r11, r12, r13, r14, r17], while showcasing a practical architectural integration within a modern simulation platform.

The proposed framework is modular and extensible, providing a foundation for incorporating V2I communication, mixed-traffic modelling, deep reinforcement learning policies, and advanced security mechanisms. As autonomous driving technology continues to evolve, such integrated simulation frameworks will be critical for developing, testing, and validating safe, cooperative, and secure intelligent transportation systems.

10 ACKNOWLEDGMENT

The authors express their sincere gratitude to Asst. Prof. Muniraju M, Department of Artificial Intelligence and Machine Learning, SJC Institute of Technology, for his continuous guidance and support throughout the project. The authors also thank the management and staff of SJCIT for providing the necessary infrastructure and facilities.

References

- [r1] S. Tang, Z. Zhang, Y. Zhang *et al.*, “A survey on automated driving system testing: Landscapes and trends,” *ACM Transactions on Software Engineering and Methodology*, vol. 32, no. 5, 2023.
- [r2] K. Muhammad, A. Ullah, J. Lloret, J. Del Ser, and V. H. C. de Albuquerque, “Deep learning for safe autonomous driving: Current challenges and future directions,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 7, pp. 4316–4336, 2021.

- [r3] Y. Wang, Z. Han, Y. Xing, S. Xu, and J. Wang, "A survey on datasets for the decision making of autonomous vehicles," *IEEE Intelligent Transportation Systems Magazine*, vol. 16, no. 2, pp. 24–39, 2024.
- [r4] T. Zhang, H. Liu, W. Wang, and X. Wang, "Virtual tools for testing autonomous driving: A survey and benchmark of simulators, datasets, and competitions," *Electronics*, vol. 13, no. 17, 2024.
- [r5] Q. Li, Z. Peng, L. Feng, Q. Zhang, Z. Xue, and B. Zhou, "MetaDrive: Composing diverse driving scenarios for generalizable reinforcement learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 3, pp. 3461–3475, 2023.
- [r6] J. Dinneweth, A. Boubezoul, R. Mandiau, and S. Espié, "Multi-agent reinforcement learning for autonomous vehicles: A survey," *Autonomous Intelligent Systems*, vol. 2, no. 27, 2022.
- [r7] D. Tao, "Multi-Agents reinforcement learning: Technical analysis and optimization," in *Proc. 3rd Int. Conf. on Image, Algorithms, and Artificial Intelligence (ICIAAI)*, 2025.
- [r8] R. Audinys, Ž. Šlikas, J. Radkevičius, M. Šutas, and A. Ostreika, "Deep reinforcement learning for a self-driving vehicle operating solely on visual information," *Electronics*, vol. 14, no. 8, 2025.
- [r9] Y. Cao, W. ShangGuan, B. Cai, L. Chai, and W. Qiu, "Predictive trajectory planning for on-road autonomous vehicles based on a spatiotemporal risk field," *IEEE Intelligent Transportation Systems Magazine*, vol. 15, no. 1, pp. 400–413, 2023.
- [r10] L. Bai, F. Zheng, K. Hou, X. Liu, L. Lu, and C. Liu, "Longitudinal control of automated vehicles: A novel approach by integrating deep reinforcement learning with intelligent driver model," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 8, pp. 11014–11027, 2024.
- [r11] K. Rashid, Y. Saeed, A. Ali, F. Jamil, R. Alkanhel, and A. Muthanna, "An adaptive real-time malicious node detection framework using machine learning in vehicular ad-hoc networks (VANETs)," *Sensors*, vol. 23, no. 25, 2023.
- [r12] E. Ezizama, K. Tepe, A. Balador, K. S. Nwizege, and L. M. S. Jaimes, "Malicious node detection in vehicular ad-hoc network using machine learning and deep learning," in *Proc. IEEE Conf.*, 2018.
- [r13] T. Rout, S. K. Balabantaray, S. S. Jena, and A. K. Acharya, "Machine learning approach in malicious node detection," *International Journal of Research in Engineering and Science*, vol. 7, no. 9, pp. 13–22, 2019.
- [r14] R. Ezumalai and D. Santhakumar, "Real-time detection of malicious nodes using Poplar optimized sparsity-aware network for improving VANET security," in *Proc. 3rd Int. Conf. on Communication, Security, and Artificial Intelligence (ICCSAI)*, 2025.
- [r15] S. Gyawali, S. Xu, Y. Qian, and R. Q. Hu, "Challenges and solutions for cellular based V2X communications," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 1, pp. 222–255, 2021.
- [r16] A. Kanavos, D. Fragkos, and A. Kaloxylos, "V2X communication over cellular networks: Capabilities and challenges," *Telecom*, vol. 2, no. 1, pp. 1–26, 2021.

- [r17] K. Rashid, Y. Saeed, A. Ali, F. Jamil, R. Alkanhel, and A. Muthanna, “An adaptive real-time malicious node detection framework using machine learning in vehicular ad-hoc networks (VANETs),” *Sensors*, vol. 23, no. 5, 2023.
- [r18] T. Suzuki, K. Matsuda, K. Kumagae *et al.*, “Acquisition of cooperative control of multiple vehicles through reinforcement learning utilizing vehicle-to-vehicle communication and map information,” *Journal of Robotics and Mechatronics*, vol. 36, no. 3, pp. 642–653, 2024.
- [r19] P. Ren, M. Li, and Z. Su, “Research on hybrid path planning algorithm based on anti-collision,” *Journal / Proc.*, 2021.
- [r20] M. Treiber and A. Kesting, *Traffic Flow Dynamics: Data, Models and Simulation*. Springer, 2013.
- [r21] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002.
- [r22] N. Juneja and A. Grover, “A survey of V2V and V2X communication for connected vehicles,” *Journal / arXiv*, 2020.
- [r23] S. Shalev-Shwartz, S. Shammah, and A. Shashua, “Safe, multi-agent reinforcement learning for autonomous driving,” *arXiv preprint arXiv:1610.03295*, 2016.
- [r24] Z. Feng, Y. Hu, H. Zhao, and H. Yu, “Deep reinforcement learning based path planning for autonomous vehicles: A survey,” *arXiv preprint*, 2020.